

Review article

Hydroelastic slamming

ODD M. FALTINSEN

Department of Marine Hydrodynamics, Norwegian University of Science and Technology, N-7491 Trondheim, Norway

Abstract An overview of the many water-impact (slamming) problems in ship and ocean engineering is given. Theoretical and experimental drop tests of horizontal and nearly horizontal elastic plates are reviewed. It is shown that maximum pressure cannot be used to estimate maximum slamming-induced stresses when maximum pressure is large, because dynamic hydroelastic effects then become important. Further, the significance of hydroelasticity increases with decreasing dead-rise angle, increasing impact velocity, and increasing the value of the highest local natural period of the structure. It is emphasized that the slamming problem must be hydrodynamically studied from a structural point of view. Comparisons between theory and full-scale measurements of slamming-induced local strains in the wet-deck of a catamaran are presented. The importance of the rigid body vertical accelerations and the influence of the side-hulls on the impact velocity are pointed out.

Key words Slamming · Hydroelasticity · Wet-deck slamming

1 Introduction

Different physical effects occur during slamming. When the local angle between the water surface and the body surface is small at the impact position, an air cushion may be formed between the body and the water. Compressibility influences the air flow. The air flow interacts with the water flow, which is influenced by the compressibility of the water. When the air cushion collapses, air bubbles are formed. The large loads that can occur during impact when the angle between the water surface and body surface is small can cause important local dynamic hydroelastic effects. The vibrations can lead to subsequent cavitation and ventilation. These physical effects have different time scales. The important time

scale from a structural point of view is when maximum stresses occur. This scale is given by the highest wet natural period (T_{n1}) for the local structure. Compressibility and the formation and collapse of an air cushion are important initially, and normally in a time scale smaller than the time scale for local maximum stresses occur. Hence, the effect on maximum local stress is generally small. Modeling compressibility during slamming is described by Korobkin.¹

There is broad range of slamming phenomena happening in ship and ocean engineering.² Slamming on ships is often categorised as bottom, bow flare, bow stem, or wet-deck slamming. The wet-deck is the lowest part of the cross-structure connecting the two side-hulls of a catamaran. Wet-deck slamming has similarities with bottom slamming or water impact on other nearly horizontal parts of a ship, such as large overhanging sterns. Bow-stem slamming and green-water impact on deck structures are of concern for floating production storage and offloading (FPSO) units. Shipping water on deck is often associated with the “dam-breaking” problem.^{3,4} The relative vertical motions between the ship and the water can cause a vertical wall of water around the bow of the ship. In this case, the subsequent water flow on the deck has similarities with the dam-breaking flow. The consequence is water moving with high velocities on the deck. However, plunging breakers hitting the superstructure have also been observed experimentally.

Wave-induced ship motions can cause violent fluid motion (sloshing) in a partially filled tank when either surge, sway, roll, or pitch motions contain energy in a frequency domain close to the lowest natural frequency for the fluid motion in the tank. This is not uncommon in sea states in which ships operate. The violent fluid motion causes large slamming loads in the tank, and may also affect the global ship motions. Water impact against a horizontal tank roof during sloshing has similarities with drop tests of horizontal elastic plates. The

Address correspondence to: O.M. Faltinsen
Received: October 4, 2000

tank roof impact will also cause hydroelastic vibrations in the tank wall adjacent to the impact area. Faltinsen⁵ studied this using a hydroelastic beam theory. The effect of a chamfered tank roof was also investigated. This problem is similar to water entry of a wedge before the horizontal part of the roof is reached. The effect of hydroelasticity decreases with an increasing dead-rise angle of the wedge. The tank-roof impact also causes damping of the fluid motion. Faltinsen and Rognbakke^{6,7} and Rognbakke and Faltinsen⁸ estimated the damping by assuming that the kinetic and potential energies in the jet caused by the impact were dissipated when the jet hits the free surface. This process resembles rainfall on water. The ambient sloshing flow was described by the analytically based multimodal methods of Faltinsen et al.⁹ and Faltinsen and Timokha.¹⁰ A rectangular smooth tank is assumed. These methods are computationally efficient and accurate relative to a direct numerical method. The structure of multimodal models facilitates coupling with the global ship motions.

Secondary impact on ship hulls due to flow separation should also be considered. An example is water entry of a cross section with a sonar dome. The jet flow separating from the sonar dome can cause secondary impact on the hull. Water impact on the air bags of an SES (air-cushion-supported catamaran) can effect the cobblestone oscillations, i.e., the wave-excited resonance oscillations of the air cushion, causing important vertical accelerations of the vessel in small sea states.^{11–14} The air bags which enclose the aft end of the air cushion have a very small clearance from the water in a calm sea. Cobblestone oscillations are a passenger comfort problem and not a structural safety problem. The behaviour of a planing bag bouncing on a free water surface was analyzed as a water entry problem, assuming a large forward speed of the SES relative to the relative vertical velocity between the bag structure and the water surface. Gravity can be neglected. Structural membrane effects are dominant. The wetted length of the bag will vary considerably through the impact on the free water surface. The hydrodynamic loading will be much larger than the aerodynamic loading due to the relatively low pressure in the air cushion of the SES and inside the bag. This implies that the immersion of the bag is low. The body boundary conditions can therefore be transferred to a straight horizontal line. At the trailing edge (separation point), it is necessary that the flow leaves the wetted surface smoothly, and that hydrodynamic pressure is continuous and equal to the atmospheric pressure (Kutta condition). This, together with the free surface conditions, leads to a square-root singularity in the hydrodynamic pressure at the spray root of the leading edge. This is similar to a 2-D steady-planing problem. The boundary value problem for the velocity

potential in the fluid has to be solved at each time step. To find the wetted length, an integral equation is set up based on keeping track of particles on the free water surface that hit points on the upstream side of the bag at different time instants. The particles are convected in the free stream direction with the free stream velocity. The vertical displacement is found by integration over time of the vertical fluid velocity. The hydrodynamic problem is analogous to the transient oscillation of a lifting elastic foil with time-dependent length. Many modes are needed to describe the dynamic behavior of the bag during water impact. Improved convergence was achieved by introducing a bending stiffness term. The spatially varying pressure underneath the bag forces the bag to reduce the leakage height. A flexible bag will have a smaller air gap than a rigid bag. A reduced air gap means less damping of the cobblestone oscillations. An important effect of the water impact is that the bag is punched out of the water immediately after the initial impact, so that an air leakage is recovered. This implies a reduction of the cobblestone oscillations.

Arai and Miyachi¹⁵ presented a numerical hydroelastic study of water impact on cylindrical shells. The elastic deformations were expressed analytically by normal modes. The hydrodynamic pressure, which is a function of the elastic vibrations, is found by a CFD (computational fluid dynamics) method based on incompressible fluid and Euler's equation of motion with gravity. A finite difference method was used, and the uprise of the free surface was accounted for by the volume of fluid method. Arai and Miyachi¹⁵ showed good agreement between their experimental and numerical results of strains during the impact. Hydroelasticity was important. This means one cannot first obtain the hydrodynamic loading by assuming that the structure is rigid and then use this as a loading in structural response calculations. Arai and Miyachi¹⁵ also showed that the rigid-body deceleration can have an important effect on strains. This was illustrated by experimental and numerical studies of a light-weight model. Kvålsvold¹⁶ pointed out that using a direct numerical solution can easily lead to large errors. This is because many structural modes are excited initially, and high accuracy is needed in the flow calculations to predict the time rate of change of the wetted area.

There are similarities between slamming on ships and on offshore platforms. Breaking waves can impact on a ship hull or the columns of a platform. Run-up along the columns can cause local damage of the platform deck. A platform is normally designed with an air gap to avoid global water impact. However, this impact may happen on bottom-mounted platforms due to an unanticipated subsidence of the sea floor, as in the Ekofisk field. Since there is interest in increasing the weight of future and

existing floating platforms, e.g., by additional processing equipment, global deck slamming is an issue for floaters. Baarholm and Faltinsen¹⁷ reported experimental and theoretical studies of global water impact on idealised platform decks. The absolute value of the minimum force is larger than the initial impact force and occurs at maximum wetted surface. Deck-generated free surface waves can occur in this time scale. This is less pronounced for long incident waves relative to deck length. The water exit phase, i.e., when the wetted surface decreases, is complicated. The time duration is considerably longer than for the deck intersection with the incident waves. An important part of the minimum force is equal to the heave added mass of the wetted platform deck times the vertical incident wave acceleration at the average impact position.

Nonlinearities in the waves and their interaction with the structure may affect the impacting water surface. Strong nonlinearities are important when describing the ambient sloshing flow. The ambient flow during wet-deck slamming on a catamaran can be strongly dependent on the forward speed, and on the frequency-dependent nonlinear side-hull and free surface effects. This cannot be predicted sufficiently accurate by current state-of-the art computer programs. Faltinsen and Zhao¹⁸ partly analysed this problem by a 2-D analysis in a cross-sectional plane. The side hulls were wedge-shaped and could cause considerable amplification of the impact velocity. This is a simple consequence of continuity of fluid mass. Second-order wave effects at least must be included in an analysis of global water impact on platform decks. Since severe sea states are of primary interest, the incident wave field is most important. However, the hull effect and the interaction with current should also be considered.

The structural engineer often wants a design pressure from the hydrodynamicist. When the problem is strongly hydroelastic, this has no physical meaning. One can, of course, define an equivalent pressure load that causes the same maximum strain in the structure. This may be convenient, but it is artificial. The equivalent pressure load would be an order of magnitude smaller than the maximum physical pressure. Since many physical effects, such as compressibility and air cushions, will occur during slamming, the hydrodynamicist may analyse problems on a time scale that is irrelevant for maximum structural response. The problem must be analysed hydrodynamically from a structural point of view. Local hydroelastic effects have to be considered. This matters if the angle between the impacting free surface and the body is small. How small this angle should be depends on the impact velocity and the highest natural period of the local structure. This is further discussed in the main text. When analysing bow flare slamming, the structure can normally be considered as

rigid in the hydrodynamic analysis. A design pressure then has a physical meaning.

There is a tendency to apply generic 3-D CFD codes that try to include all physical effects. However, only the physical effects that matter during water entry should be used, otherwise numerical problems are likely to occur. For instance, the prediction of the intersection between the body and the free surface is often very difficult, but is important. This is particularly true when the angle between the impacting free surface and the body surface is small. Analytical studies are helpful in this context. On the other hand, the fluid can generally be assumed to be incompressible and the flow irrotational.

When the duration of the water entry loads is of the order of the global natural periods of a ship or a floating platform, they influence the global body motions and loads. Whipping, i.e., global transient resonant flexible motions, has been considered both for ships and catamarans. Local hydroelastic effects will not matter in global hydroelastic analyses. Faltinsen¹⁴ has discussed wave-induced hydroelastic effects on high-speed vessels.

The following description concentrates on local hydroelastic effects. Slamming on ship hulls, including wet-decks, is studied in detail. The presentation is biased toward the author's own work.

2 Local slamming on ship hulls

Fundamental theoretical and numerical studies of wave impact on horizontal or nearly horizontal one-beam and three-beam models will be reported first. The motivations for these studies were slamming on wet-decks with horizontal cross sections. The objective was to find the maximum strains in the longitudinal stiffeners. The analogy for the one-beam model can be described by Fig. 1. One beam is used to describe the part of the wet-deck between two transverse frames. The beam is then a two-dimensional model of one longitudinal stiffener and the adjacent plate field, which acts like a flange. Half the width of the flange is assumed to be equal to half the distance between two longitudinal stiffeners. The beam is supported by the transverse frames at the beam ends. A three-beam model means that the wet-deck between four consecutive transverse frames are modeled. In the model tests, the beam is a plate of uniform thickness. The scaling was based on using realistic structural data for wet-decks.

2.1 One-beam model

Theoretical and experimental studies of wave impact on horizontal elastic plates (one-beam models) made of

steel and aluminum have been presented by Faltinsen,¹⁹ Faltinsen et al.,²⁰ Kvålsvold,¹⁶ Kvålsvold and Faltinsen,²¹ Kvålsvold et al.,²² and Aarsnes.²³ The water was assumed to be incompressible, and the flow irrotational and two-dimensional in the theoretical studies. 2-D Euler beam theory was used. The beam deflections were represented by normal modes, and the interactive behavior of the structural vibrations and the fluid oscillations were studied. A high degree of analysis was possible. This was important for numerical stability. The original theoretical studies by Kvålsvold¹⁶ and Kvålsvold and Faltinsen²¹ incorporated the effect of axial and shear deformations and rotary inertia, but they found that Euler beam model was a satisfactory approximation. The original studies included many modes, but the experimental studies clearly showed that the lowest mode had the dominant effect on the maxi-

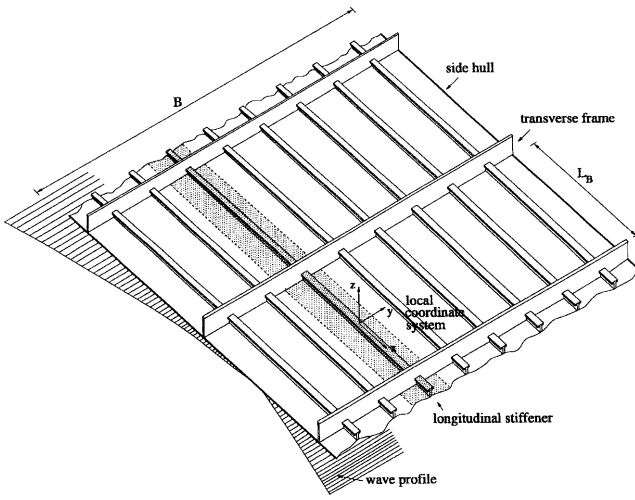


Fig. 1. Part of the wet-deck. The shaded part is included in the structural beam model. In the one-beam model, the shaded longitudinal stiffener and plate between two transverse frames are considered. In the three-beam model, the shaded area between four consecutive transverse frames is considered²⁸

mum strain values. This was a motivation for Faltinsen¹⁹ to simplify the problem significantly. Significant dynamic hydroelastic effects were demonstrated both in theory and experiments. Korobkin and Khabakhpasheva²⁴ have also done theoretical studies on the experiments of Aarsnes.²³

The main parameters for the plates used in the drop tests are given in Table 1. The shape of the plate is shown in Fig. 2. The total drop section was divided into three parts, one measuring section with a dummy section on each side, as shown in the figure. The measuring section was connected to the drop rig using two force transducers. The total weight of the drop rig was 500 kg.

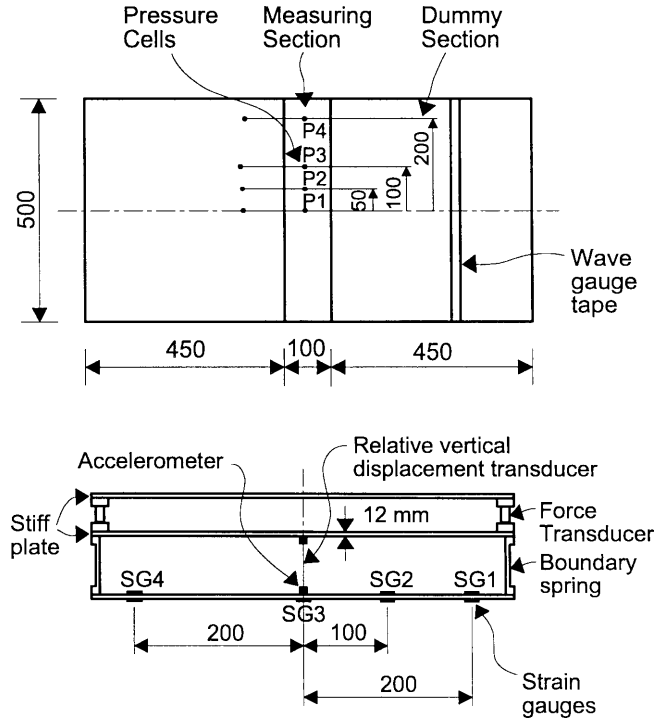


Fig. 2. Details of the elastic test plate, one-beam model. Length scale in mm²³

Table 1. Main parameter for plates used in the drop tests of a one-beam model

Parameter	Plate I (steel)	Plate II (aluminum)
Structural mass per unit length and breadth, M_B	62 kg/m ²	21 kg/m ²
Modulus of elasticity, E	210×10^9 N/m ²	70×10^9 N/m ²
Length of plate, L	0.50 m	0.50 m
Breadth of plate, B	0.10 m	0.10 m
Bending stiffness, EI	8960 Nm ² /m	17060 Nm ² /m
Structural mass parameter, $M_B/\rho L$	0.124	0.042
Connecting spring parameter, $k_s L/(2EI)$	2.85	1.50
Distance from neutral axis to strain measurements, z_a	0.004 m	0.01375 m

This large weight was used to ensure small variations in the drop speed during impact. Before the drop tests, the end connecting moment of the plate was measured. This can be expressed as $-k_\theta \theta_{be}$, where θ_{be} is the rotation angle and k_θ is the spring stiffness as shown in Table 1.

The instrumentation in the test consisted of pressure cells, vertical force transducers connected to the measuring section, wetted-surface measurements using wave gauge tape, accelerometers to determine the accelerations in the rig and the measuring plate, drop velocity measurements, strain measurements using strain gauges in four different positions on the plate, and the vertical deflections of the plates using a displacement transducer and wave staffs. The tests were carried out for drop speeds ranging from 2.2 m/s to 6.2 m/s.

Measured nondimensional maximum strains in the middle of the plate are shown in Fig. 3 as a function of the nondimensional impact velocity V . The scaling assumes that the maximum strain is proportional to V , and that T_{n1} is the time scale. T_{n1} is the highest wet natural period for the test plate. The plates have no forward speed. ρ is the mass density of the fluid. The impact position and radius (R) of curvature of the waves at the impact position were varied. Incident regular waves as well as calm water ($R = \infty$) were used. A wave crest was intended to initially hit between the plate ends in all cases shown in Fig. 3. The experimental results decrease slightly with decreasing nondimensional impact velocity. If the largest L/R value is disregarded, the maximum strain shows little influence from L/R . The largest L/R value is normally unrealistic for wet-deck slamming. This can be found by using realistic values of L and noting which wave conditions cause large vertical relative motions of a multihull vessel. Table 1 shows

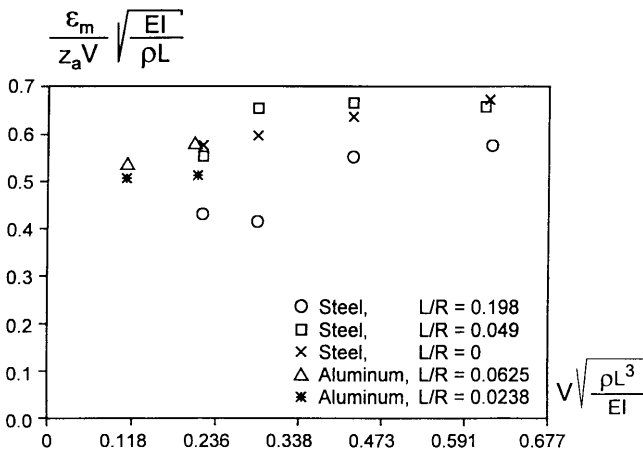


Fig. 3. Measured maximum nondimensional strain amplitude ε_m in the centers of the horizontal steel and aluminum plates as a function of the nondimensional impact velocity. One-beam model

that the steel and aluminum plates have different structural mass and connecting spring parameters, but this difference is insignificant.¹⁹ The asymptotic theory of Faltinsen¹⁹ agrees well with the experiments. This theory gives a nondimensional maximum strain for a given structural mass and connecting spring parameter that is independent of impact speed and wave characteristics.

The physics can be explained as follows. The plate experiences a large force impulse during a small time relative to the highest natural period for the plate vibrations. This is called the structural inertia phase, which implies that structural elasticity is unimportant and the impact force is balanced by structural inertia forces. The force impulse causes the space-averaged relative velocity between the elastic vibration velocity and V to be zero at the end of the structural inertia phase. The plate then starts to vibrate as a free vibration with an initial vibration velocity V and zero deflection. Maximum strains occur during the free vibration phase. Only the lowest eigenmode is important in the time scale of when maximum strains occur. The details of the pressure distribution during the first initial phase are not important. Very large pressures that are sensitive to small changes in the physical conditions may occur in this phase. This can be seen from the collection of measured maximum pressures during the tests (Fig. 4). Figure 5 gives an example of the time history of measured pressures at points P1 and P3 (see Fig. 2). The time duration of large pressures is very small. For P1 it is only 10^{-5} s. The maximum strains occur much later, i.e., after about 5×10^{-3} s. Figure 5 also shows that the large pressures are very concentrated in space. The measured maximum strains showed a very small scatter for given impact velocity and plate even when maximum pressure varied

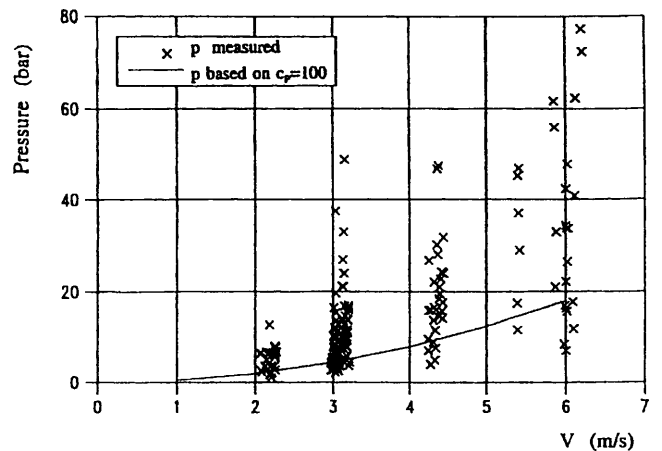


Fig. 4. Measured maximum pressure from different drop tests of horizontal plates as a function of the drop speed. One-beam model. $C_p = p / (0.5 \rho V^2)$ is the pressure coefficient

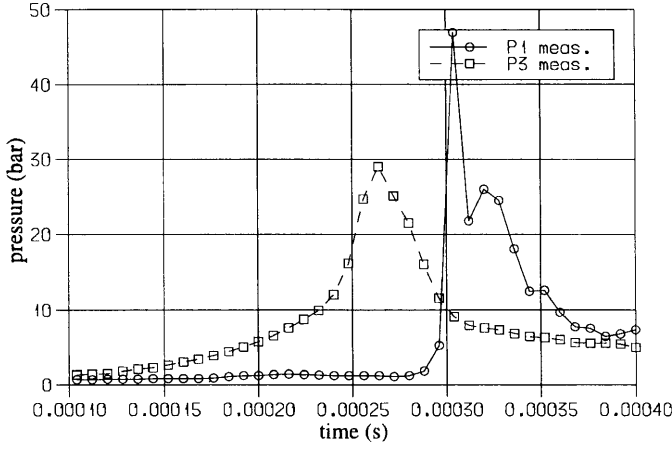


Fig. 5. An example of measured pressures during the impact of the elastic plate shown in Fig. 2. Drop height is 1.8m ²³

widely. The largest measured pressure was approximately 80 bar for $V = 6.2\text{m/s}$. The diameter of each pressure cell was 4mm. Smaller pressure cells may have resulted in even larger pressures. A sampling frequency up to 500kHz was used. These results show that it is misleading from a structural point of view to measure pressures when hydroelasticity is important.

The behaviour of a possible air cushion during the initial structural inertia phase was explained in a simplistic way by Faltinsen.¹⁹ The air cushion is a consequence of the air flow between the body and the impacting water surface. When the air cushion is created, the time-dependent air cushion pressure $p_{ac}(t)$ is assumed to be space-independent and related to the air cushion mass density by an isentropic air compression relation. Space variations in p_{ac} may matter in reality. The continuity equation for the air cushion was used. Since the ratio between the structural mass density and fluid mass density is small, the volume change of the air cushion due to free-surface deformation can be neglected relative to the effect of the elastic beam deformation w . Finally, the structural inertia force is proportional to p_{ac} . The time scale T_{ac} of the air cushion process follows from the equations. By using measured data by Miyamoto and Tanizawa²⁵ for the thickness of the air layer and the air cushion pressure for a rigid plate, it follows that T_{ac} is small relative to the time scale of the free vibration phase. When the air cushion collapses, the beam has asymptotically zero deflection and a space-averaged deflection velocity $d\bar{w}/dt$ which is equal to the rigid body drop velocity V . This means that on average, the beam has zero velocity. The same result was obtained if no air cushion was initially created.

Faltinsen¹⁹ also studied the free vibration phase when the forward speed U is large relative to the drop velocity. The behaviour of the elastic beam is similar to linear steady-state vibrations of a lifting elastic foil. When

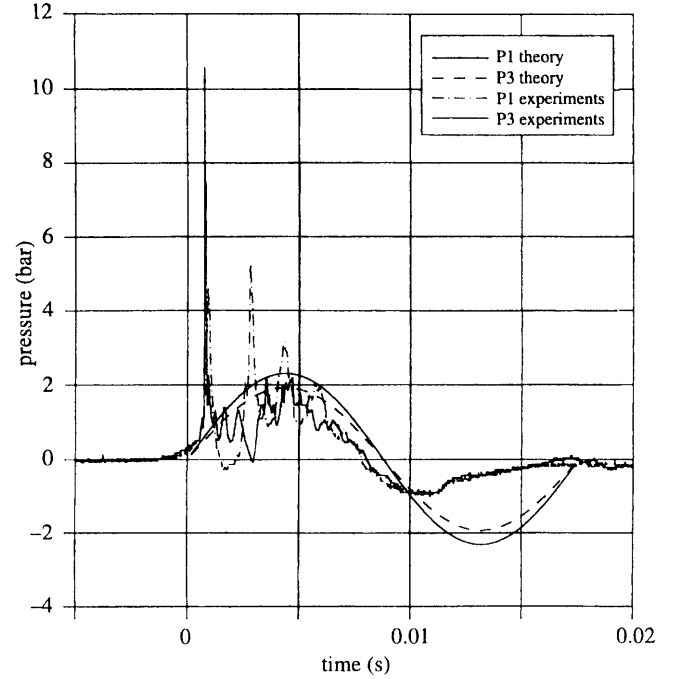


Fig. 6. Pressure at two positions of a dropped horizontal elastic plate as a function of time. Comparison between asymptotic theory and drop tests¹⁹

$T_{n1}U > \sim L$, where T_{n1} is the highest wet natural period of the beam, the forward speed can be accounted for simply as an angle of attack effect in the body boundary condition. This is satisfied for realistic material properties and existing commercial vessels. This means that the maximum strains are proportional to $V_r + U\alpha$, where α is a small rigid body rotation in radians. $\alpha = 0$ means a horizontal beam. A positive α means that the trailing edge is downwards relative to $\alpha = 0$. V_r can be interpreted as the drop velocity relative to the vertical ambient wave velocity u_z , but u_z is negligible relative to the drop velocity in the cases studied previously. The consequence is that V occurring in the nondimensional strain values presented in Fig. 3 can be replaced by the effective drop velocity

$$V_T = V + U\alpha + u_z \quad (1)$$

This was documented experimentally for a three-beam model and will be discussed later in the text.

Figure 6 shows an example from Faltinsen¹⁹ on the time history of the measured and theoretical pressures on a dropped horizontal elastic plate of length 0.5m. The forward speed is zero. P1 and P3 represent the pressure in the centre of the plate and at a position 100mm from the centre, respectively. Atmospheric pressure is not included. The theoretical pressures describe only the free vibration phase. A qualitative understanding of the theoretical pressure can be obtained

by considering a plate that is forced to oscillate with a velocity equal to the space-averaged deflection velocity $d\bar{w}/dt$. The oscillation period is equal to the highest wet natural period T_{n1} of the beam vibrations. The pressure is in phase with $-d^2\bar{w}/dt^2$ and gives the added mass force when integrated over the body. This means that positive and negative pressures occur during the first and second half of the oscillation period, respectively.

The figure shows that large underpressures relative to atmospheric pressure are present in the second half of the oscillation cycle. The pressures at P1 and P3 are close to the vapour pressure around time equal to 0.01 s both in theory and experiments. This means that cavitation occurs. Owing to the large underpressures under the plate during the second half of the first complete oscillation and the low submergence of the plate, air will be drawn in under the plate and the pressure will become close to atmospheric (see time longer than 0.015 s in Fig. 6). This means ventilation. This effect is also evident in the measurements of the wetted length, which shows a significant reduction during this time interval. The consequence of this reduction in the wetted length due to ventilation and cavitation is a pronounced reduction in the added mass, and hence a reduction in the oscillation period. This causes a lower strain amplitude. In fact, the oscillation period for time longer than about 0.014 s is very close to the highest natural period of the plate in air. These effects are not included in the theory. However, the maximum strains occur during the first half of the oscillation cycle. The numerical and experimental strain values are in good agreement for this part of the impact. When it comes to pressures, pronounced peaks are present in the experimental results in the initial phase. These peaks are often the only things that are analysed, and they are then referred to as the slamming pressures. The asymptotic theory of the free vibration phase does not include these effects. By disregarding the experimental peaks in Fig. 6, there is reasonable agreement between theory and experiments up to cavitation occurring at around time equal to 0.01 s.

The measured maximum pressures in P1 and P3 differ in magnitude and time history. If an air cushion is present when maximum pressures occur, this contradicts a simplistic air cushion theory, which assumes space-independent pressure. If we postulate instead that an air cushion is present with spatially varying pressures, the propagation speed of the pressure signal should be associated with the speed of sound in air. However, if the maximum pressures in P1 and P3 are examined on a very small time scale, as in Fig. 5, the propagation speed of the pressure signal is clearly higher than the speed of sound. One explanation may be that the pressure behavior is associated with a very rapid change in the wetted surface, starting from the

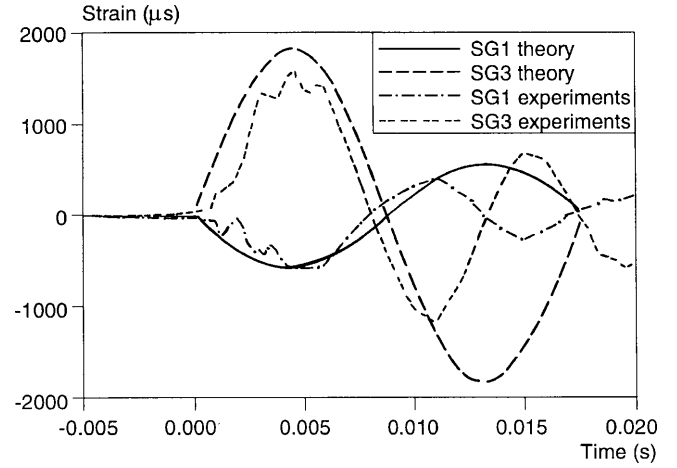


Fig. 7. The strains at different locations along the beam as a function of time (see Fig. 2). Comparison between asymptotic theory and drop tests. Drop height is 0.5 m¹⁹

ends of the beam. Let us assume for the moment that the fluid may be considered incompressible and that a jet flow is present at the intersection between the free surface and the beam. If we make the analogy with the jet flow studied by Wagner²⁶ (see also Faltinsen²⁷) during the impact of a blunt 2-D body, the maximum pressure would be $0.5\rho(dc/dt)^2$ where dc/dt is the velocity of the spray root. However, this is speculative and would require more detailed experimental studies to justify. It is likely that the compressibility of the water matters during this phase. Further, the initially created high pressures have a damped oscillatory behavior over a time scale that is not small relative to T_{n1} . However, the reported good agreement between experimental and theoretical maximum strains indicates that the large pressure oscillations are not important from a structural point of view.

Figure 7 shows the strains at two different locations along the beam as a function of time. Strains at other positions were also studied with equally good agreement between theory and experiments. A drop velocity of 2.5 m/s is used in the theoretical results. This is based on an examination of the experimental drop velocity $V(t)$ and the use of a representative value after the structural inertia phase. Only the lowest structural mode is used in the calculations. The reasons for doing this are that there is a clear difference in the natural periods of the different modes, and that the influence of higher modes is small on the time scale of when maximum strains occur. The latter is due to both structural damping and initially small modal amplitudes. Figure 7 shows that initially the strains are equal to zero and increase to a maximum value after 4–5 ms. The theoretical oscillation period is approximately 17 ms. There is good agreement between the experimental and numeri-

cal values for the maximum (or minimum) strains and the oscillation period during the first half of the oscillation period. However, there is a significant difference between the theoretical and experimental values for the strains and the oscillation period during the second half of the oscillation period. Since the oscillation period is significantly reduced, this effect cannot be explained by damping. The difference can be explained by the occurrence of cavitation and ventilation in the tests, and the fact that the structure oscillates as if it is in air.

Since gravity does not matter in the previous studies of hydroelasticity, the orientation of the body does not need to be horizontal. What is important is that there is a small angle between the impacting free surface and the body surface. An example could be a vertical wave front hitting a vertical wall.

2.2 Three-beam model

Haugen²⁸ (1999), Haugen and Faltinsen²⁹, and Haugen et al.³⁰ studied theoretically the hydroelastic impact of plates with three beam elements. The physics in the initial impact phase of three beam elements is somewhat different from that of one beam element. The wetting of the plates lasts relatively longer and air cushion effects are believed to matter more. Further, there is one dominant high natural period for one beam element, while there are several for three beam elements. The final results in terms of maximum strains are similar, but somewhat higher for three beams than for one beam. The theoretical results were compared with experimental results by Aarsnes.³¹ The drop section was similar to those for the one-beam model (see Fig. 2) divided into three parts, one measuring section with a dummy section on each side. The plates were made from HSLA (high strength low alloy) steel material. The plate thickness was 2.5 mm and the distance from the neutral axis to the stress measurements z_a was 1.25 mm. The bending stiffness EI was 273 Nm²/m. The length L_B of each plate (beam) was 0.2 m. The thickness and stiffness of the dummy section plates were the same as for the measuring section. The measuring section was connected to the rig by four force transducers. The elastic plates were connected to the force transducers through the boundary springs, as shown in Fig. 8. The total weight of the measuring section includes the weight of the test plate in addition to the boundary springs, i.e., it represents the total weight of the section connected underneath the vertical force transducers. This weight was 2.21 kg. The total weight of the drop rig, including ballast weight, was 395 kg. The structural mass parameter of the plates was $M_B/\rho L_B = 0.098$. Horizontal plates with zero forward speed, and plates mounted with a constant pitch angle $\eta_5 = 5.8^\circ$ with forward speed $U = 3.5$ m/s were tested. The drop speed V varied from

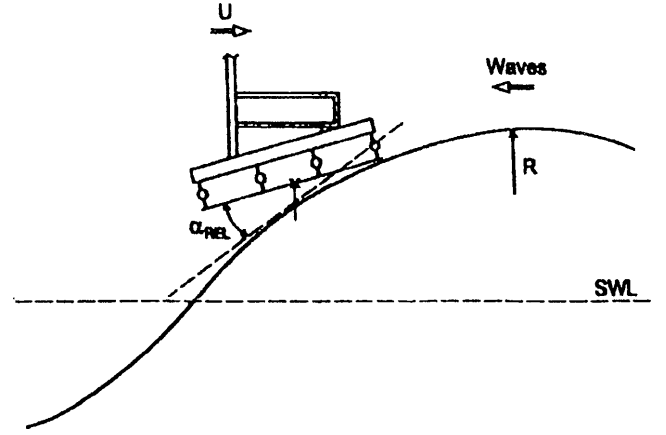


Fig. 8. The shape of the plates seen from the side for the three-beam model. Definition of the relative angle α_{rel} (see Eq. 2)

1.0 m/s to 3.2 m/s. Five different regular wave conditions were used. The radius of curvature of the wave crest, R , for the different waves varied from 3.86 m to 15.1 m. Drop against calm water (i.e., $R = \infty$) was also included. In most tests, the wave crest was intended to hit the center of the plate. However, in some tests the position where the wave crest hits the plate was also systematically varied. The end connecting moment of the plate was determined prior to the drop tests. The resulting connecting spring parameters were $k_\theta L_B/2EI = 3.65$.

The relative angle between the plate and the wave surface at the time of initial impact is defined by

$$\alpha_{rel} = \eta_5 - k\zeta_A \sin(kX_T) \quad (2)$$

where ζ_A is the wave amplitude and k is the wave number of the regular waves (Fig. 8). X_T is the distance from the wave crest to the center of the plate. According to this definition, $\alpha_{rel} = 0$ means that the wave initially hits in the center of the plate, and that the plate and the wave profile are parallel. $\alpha_{rel} \approx 0$ means that the wave initially hits close to the center of the plate, and it is important to note that the plate and the wave profile are still parallel. We further define an effective drop velocity as in Eq. 1. The maximum strain amplitude ϵ_m will be nondimensionalized, as in Fig. 3, but with V_T as drop velocity instead of V and with the length L_B of each plate (beam) instead of L . The influence on the strains where the waves initially hit the plate model was investigated experimentally by varying the relative angle between the plate and the wave surface at the moment of initial impact. The measured maximum nondimensionalized strain amplitudes in the centers of the plates are shown in Fig. 9 as function of α_{rel} . The forward speed is zero in this case. The results are shown for different values of the wave crest radius, R . The influence of varying the wave crest radius is found to be very small.

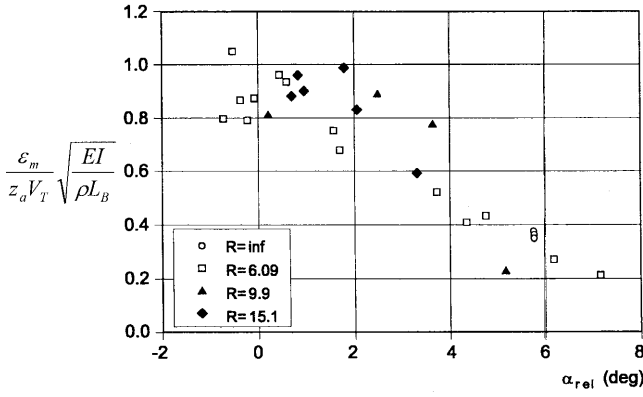


Fig. 9. Measured maximum nondimensional strain amplitude ε_m in the center of the plate model as function of α_{rel} for the three-beam model. Forward speed is zero³¹

This is consistent with the previous results for a one-beam model. For $|\alpha_{rel}| < 2^\circ$, the stresses are approximately independent of the relative angle, i.e., of the position of the wave crest relative to the center of the plate model. For larger values of $|\alpha_{rel}|$, the maximum nondimensionalized strains decreases significantly with increasing $|\alpha_{rel}|$. When $|\alpha_{rel}| > \sim 2^\circ$, one of the edges of the plate penetrates the free surface first. The reason why the strains decrease with increasing $|\alpha_{rel}|$ is that the ratio between the hydrodynamic loading time and a characteristic value for the highest natural period of the elastic vibrations increases. We will come back to this point later when we discuss water entry of a wedge. The theoretical predictions by Haugen²⁸ gave satisfactory results when $|\alpha_{rel}| > \sim 2^\circ$. The less satisfactory results for small $|\alpha_{rel}|$ were explained by Haugen²⁸ as air-cushion effects which last longer for a three-beam model than for a one-beam model. The effect of forward velocity on the resulting stresses and deflections of the plate was also evaluated. Measured maximum nondimensionalized strain amplitudes in the center of the plate model are shown in Fig. 10. Results are shown only for tests where the wave hits parallel to the plate. The wave crest radius varies, but this has no influence on the results. The results owing to different forward velocities may indicate a small influence of V_T on the measured nondimensionalized strains in terms of a small increase for increasing V_T . However, as a first approximation, it may be stated that there is no variation in the results with respect to V_T . Since the maximum strain amplitude is made nondimensional with respect to V_T , the maximum stress amplitude is proportional to V_T .

2.3 Relative importance of hydroelasticity

Faltinsen³² studied the relative importance of hydroelasticity for an elastic hull with wedge-shaped cross

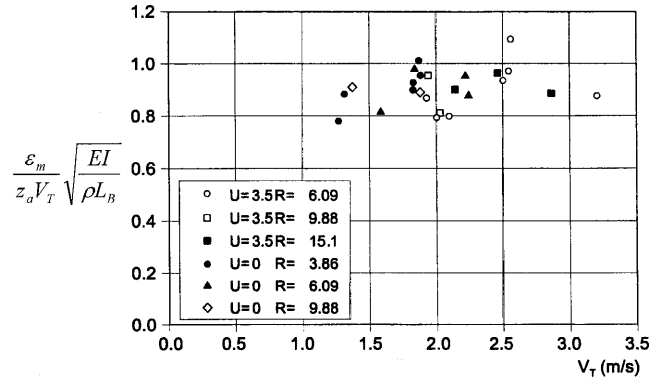


Fig. 10. Measured maximum nondimensional strain amplitude ε_m in the center of the plate model as function of the total relative normal velocity V_T for the three-beam model. The effect of forward speed U . α_{rel} is less than 2° ³¹

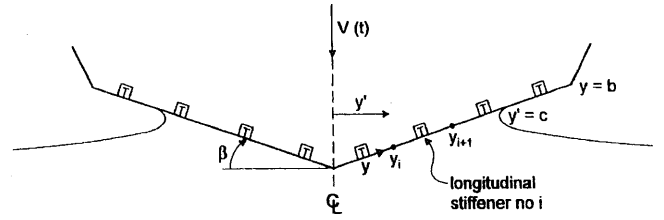


Fig. 11. Water entry of a wedge-shaped elastic cross section

sections penetrating an initially calm water surface (Fig. 11). The main application was a wet-deck with nonhorizontal crosssections, but since the studies are fundamental, they have a broader application. A stiffened plating between two rigid transverse stiffeners or frames was examined (Fig. 12). A hydrodynamic strip theory in combination with orthotropic plate theory was used.

The differential equation for the lateral deflection w of the plate is written as

$$\bar{m} \frac{\partial^2 w}{\partial t^2} + D_x \frac{\partial^4 w}{\partial x^4} + 2B \frac{\partial^4 w}{\partial x^2 \partial y^2} + D_y \frac{\partial^4 w}{\partial y^4} = p(x, y, t; w) \quad (3)$$

Here p is the hydrodynamic pressure, which is a function of time t and space and depends on w . The x - y plane is parallel to the plate plane. The x -direction is in the longitudinal direction of the hull. The x -coordinates of the two transverse girders are $x = 0$ and L . The y -coordinate of the keel is $y = 0$. The length of the plate in the y -direction is b . $\bar{m}$ is the average structural mass of the plate per unit area. The flexural rigidities D_x and D_y and the effective torsional rigidity B can be expressed as

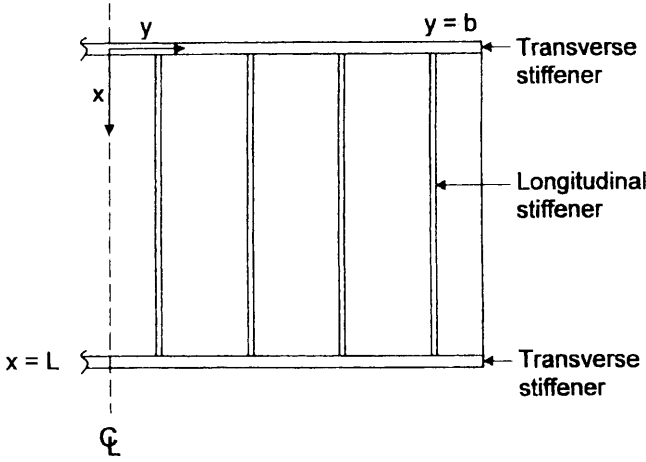


Fig. 12. Stiffened plating consisting of plate and longitudinal stiffeners

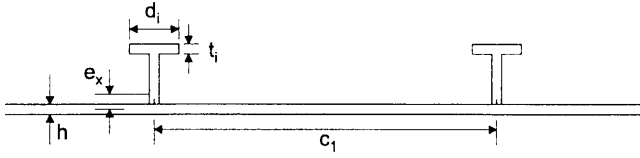


Fig. 13. Cross section of longitudinal stiffeners

$$D_x = \frac{Eh^3}{12(1-\nu^2)} + \frac{Ehe_x^2}{1-\nu^2} + \frac{EI_{0x}}{c_1} \quad (4)$$

$$D_y = \frac{Eh^3}{12(1-\nu^2)} \quad (5)$$

$$B = \frac{Eh^3}{12(1-\nu^2)} + \frac{E}{12(1+\nu)} \frac{\sum d_i t_i^3}{c_1} \quad (6)$$

where E is Young's modulus, ν is the Poisson ratio, and I_{0x} is the moment of inertia of the stiffeners in the x -direction with respect to the neutral axis. The other parameters are illustrated in Fig. 13. e_x is the distance from the middle of the plate to the neutral axis, h is the plate thickness, and c_1 is the distance between two longitudinal stiffeners. d_i and t_i are the cross-sectional length and thickness of the individual parts of a longitudinal stiffener, respectively.

The solution of Eq. 3 is based on the Vlasov–Galerkin method (see Szilard³³). The lateral deflections are expressed as

$$w = \sum_m \sum_n W_{mn}(t) X_m(x) \Psi_n(y) \quad (7)$$

where X_m and Ψ_n are eigenfunctions of vibrating beams. Clamped end conditions are assumed at $y = 0$ and $y = b$. Since the pressure loading is assumed to be symmetric about $x = L/2$, it is sufficient to consider eigenfunctions

X_m that are symmetric with respect to $x = L/2$. General end conditions were assumed for X_m . Owing to the nature of the loading, at least two modes are necessary in the y -direction. Ψ_1 is symmetric about $y = 0.5b$ and Ψ_2 is antisymmetric about $y = 0.5b$. In practise, one eigenfunction X_1 and two eigenfunctions Ψ_1 and Ψ_2 were used.

The hydrodynamic problem is solved by a generalization of Wagner's method by including the effect of elastic vibrations. Even if Wagner theory is approximate for large dead-rise angles, it is believed that the approximate theory demonstrates the main parameter dependence. The details of the flow at the spray roots are not included. The method is similar to that used by Kvålsvold and Faltinsen.²¹ An important difference is that approximate 3-D effects are included. This is done by assuming two-dimensional flow (i.e., the 2-D Laplace equation) at each cross section, but allowing for a different rise-up of water for each cross section. The orthotropic plate theory provides 3-D coupling between the flows at the different cross sections.

It was shown by Faltinsen³² that the importance of hydroelasticity for the local slamming-induced stresses increased with decreasing dead-rise angle β and increasing impact velocity V . The nondimensional parameter $\tan \beta / V(\rho L^3/EI)^{1/2}$ was introduced. V is then assumed to be constant. EI is the bending stiffness of the longitudinal stiffener and an effective flange. The parameter is proportional to the ratio between the wetting time of the rigid wedge and the natural period of the longitudinal stiffener. We can associate the wetting time of the wedge with the duration of the loading. If we make an analogy with a simple mechanical system consisting of a mass and a spring, we then know that the duration of the loading relative to the natural period characterises the dynamic effects. In fact, this explains why the nondimensionalized strain values given in Fig. 9 decrease with α_{rel} when $\alpha_{rel} > \sim 2^\circ$. The nondimensional results are shown in Fig. 14. This particular way of nondimensionalizing the results gives small explicit dependence on the dimensionless impact velocity

$$V_{ND} = V \sqrt{\frac{\rho L^3}{EI}} \quad (8)$$

Note that the strains are nondimensionalized in a very different way in Figs. 3 and 14. Figure 3 is based on the assumption that the strain is proportional to impact velocity V , while Fig. 14 assumes that ϵ_m is proportional to V^2 , i.e., as in quasi-steady-pressure loading. Further, $\tan \beta$ is introduced to account for the way that the loading depends on the dead-rise angle. This is based on analyzing the loading on a rigid wedge.

Figure 14 shows that hydroelastic effects become important when $\tan \beta < 0.25V(\rho L^3/EI)^{1/2}$ for the stiff-

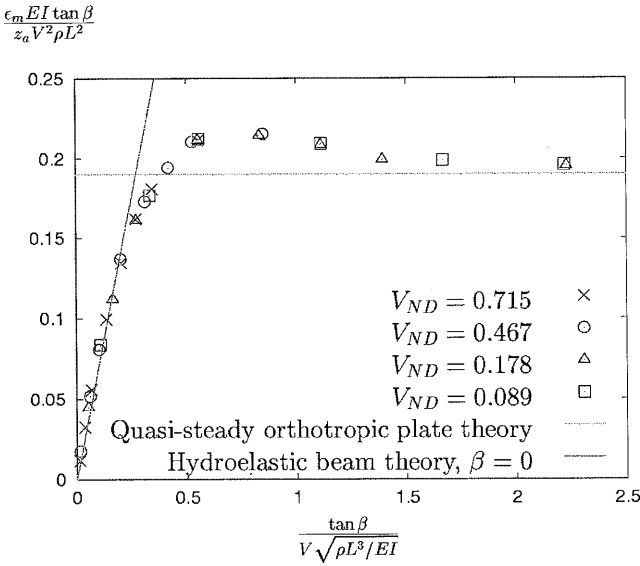


Fig. 14. Nondimensional maximum strain ε_m in the middle of the second longitudinal stiffener from the keel, presented as a function of a parameter that is proportional to the ratio between the wetting time of a rigid wedge and the natural period of a longitudinal stiffener based on beam theory. Different nondimensional constant impact velocities V_{ND} (see Eq. 8). The calculations are by hydroelastic orthotropic plate theory³²

Table 2. Main particulars for the Ulstein test vessel

Particulars	Symbols	Value	Units
Length over all	L_{0A}	~35	m
Length between perpendiculars	L_{pp}	29.8	m
Beam over all	B	9.4	m
Distance between c.l. hulls	C_{H-H}	5.6	m
Draft at FP at zero speed	T_{FP}	0.95	m
Draft at AP at zero speed	T_{AP}	1.25	m
Displacement	M	~100	ton
Longitudinal center of gravity from transom (AP)	LCG	10.9	m

c.l., center line; FP, forward perpendicular; AP, aft perpendicular

fined plating studied by Faltinsen.³² If hydroelasticity is not important, the maximum strain ε_m is proportional to V^2 . The parameter study assumed constant V during impact. In reality, V may vary substantially during the impact. This happened in the comparative studies with full-scale experiments of local slamming-induced strains in the wet-deck of a 30-m-long catamaran reported by Faltinsen.³²

2.4 Comparison with full-scale tests

The theory by Faltinsen³² was compared with full-scale tests of the Ulstein test vessel. This is a high-speed catamaran. Its main parameters are given in Table 2. Strains were measured in two areas, called positions I

Table 3. Structural data for the analyzed stiffened platings at impact positions I and II of the Ulstein test vessel

Length between transverse stiffeners (L)	1.25 m
Transverse length (b)	
Position I	1.902 m
Position II	0.951 m
Distance between longitudinal stiffeners (c_1)	0.317 m
Young's modulus for aluminium (E)	70×10^9 N/m ²
Flexural rigidity/ E ($D_x/E = I$)	8.73×10^{-7} m ³
Flexural rigidity/ E (D_y/E)	0.06×10^{-7} m ³
Effective torsional rigidity/ E (B/E)	0.06×10^{-7} m ³
Distance from neutral axis to maximum strain in longitudinal stiffeners (z_a)	0.0454 m
Structural mass density ($\bar{m}$)	18.4 kg/m ²
Spring stiffness parameter (α)	3.0
Dead-rise angle β	
Tan β (position I)	0.123
Tan β (position II)	0.25

Position I, between frames 24 and 25; position II, between frames 20 and 21

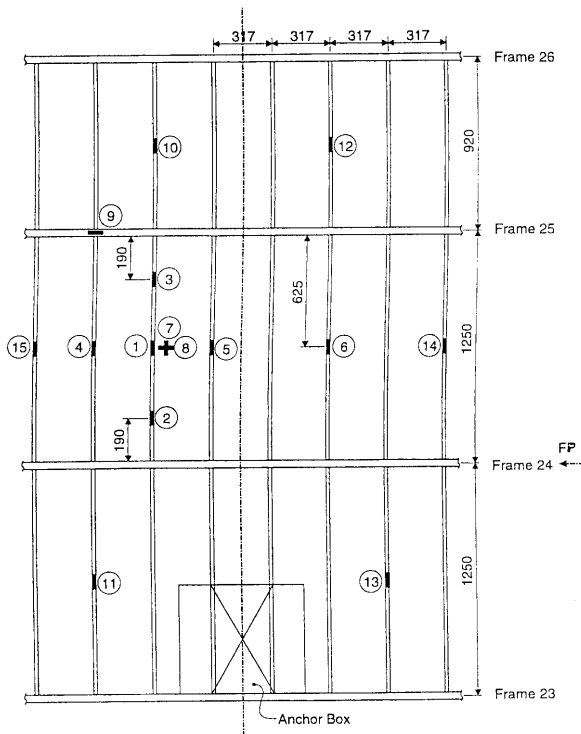
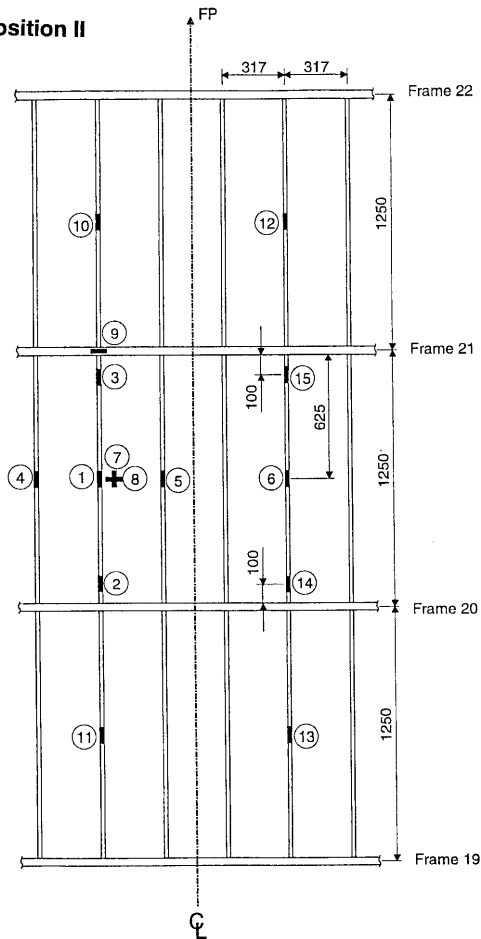
and II, in the forward part of the wet-deck (Fig. 15). Most of the strain gauges were mounted on the top of the longitudinal stiffeners. Representative cross sections of the catamaran at impact positions I and II are shown in Fig. 16. The wet-deck has a wedge-shaped cross section. The dead-rise angles are between 5° and 15° in the impact areas. Details of the measurement program were reported by Aarsnes and Hoff.³⁴

The theoretical method will first be compared with test no. 204 in the full-scale tests. The ship speed is 18 knots. The significant wave height $H_{1/3} = 1.5$ m, the peak period T_p of the wave spectrum is 6 s, and the wave direction is head sea. Eight wet-deck slams occurred during 225 heave and pitch oscillations. We will concentrate on the slam event where the highest strains were found. One difficulty is in estimating the impact velocity. Only the global ship motions were recorded, and no measurements of the wave kinematics were made. The impact velocity was estimated by approximating the incident waves by a regular wave at the time and position of impact. The influence of the side-hulls were accounted for, and this is particularly important at position II. The impacting relative velocity was expressed as

$$V = V_0 + V_1 t \quad (9)$$

in the analysis. Here $t = 0$ corresponds to initial impact time. Full details are given by Faltinsen.³²

Structural data and averaged dead-rise angles for impact positions I and II that are used in the theoretical analysis are given in Table 3. The parameter α is defined as $0.5k_\theta L/D_x$, where k_θ is the spring stiffness, which is related to the restoring moment $-k_\theta \theta_{be}$ at $x = 0$ and L . θ_{be} is the rotation angle at the ends. D_x is defined by Eq. 4. There is no exact way to estimate the spring stiffness coefficient α shown in Table 3. The value was deter-

Position I**Position II**

mined in an indirect way by examining the relative values of the measured strains along a longitudinal stiffener.

Figures 17 and 18 show the measured strains at positions I and II. The highest recorded strain was at position II.1. The average pressure over the longitudinal stiffener and the effective flange causing a steady strain equal to the maximum measured strain was only 0.2 bar. The measured maximum strains correspond to about half the yield stress. The ship was allowed to operate up to $H_{1/3} = 3.5$ m. The reason is that the classification rules wrongly specified that the wet-deck of the ship was sufficiently high above sea level to avoid wet-deck slamming. On the other hand, Aarsnes (private communication, 1997) reported that severe wet-deck slamming could be avoided by changing the ship's course. Figures 17 and 18 show only the results on one side of the center line. The results for symmetric points on the other side of the center line were almost the same. The water impacts first at position II. Figure 19 shows the theoretical predictions for position II. The numerical and experimental results are in qualitative agreement. We note that after the maximum strains have occurred at position II.1, the theoretical values at that position drop rapidly to smaller values than the experimental values. One reason for this experimental behavior is probably that the interaction with the side-hulls becomes stronger than the theory assumes. The reason why the theoretical values become negative is that the negative "added mass" pressure component associated with global ship accelerations dominates. The experimental values also become negative. The presence of large ship accelerations also has an impor-

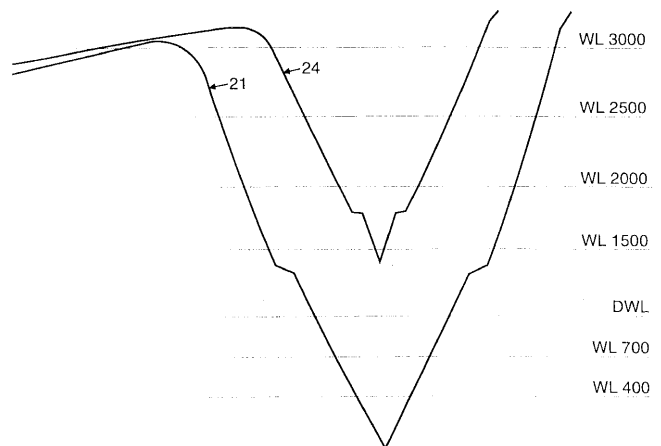


Fig. 16. A representative cross section of the Ulstein test vessel at impact positions I and II

Fig. 15. The locations of strain gauges at positions I and II in the wet-deck of the Ulstein test vessel

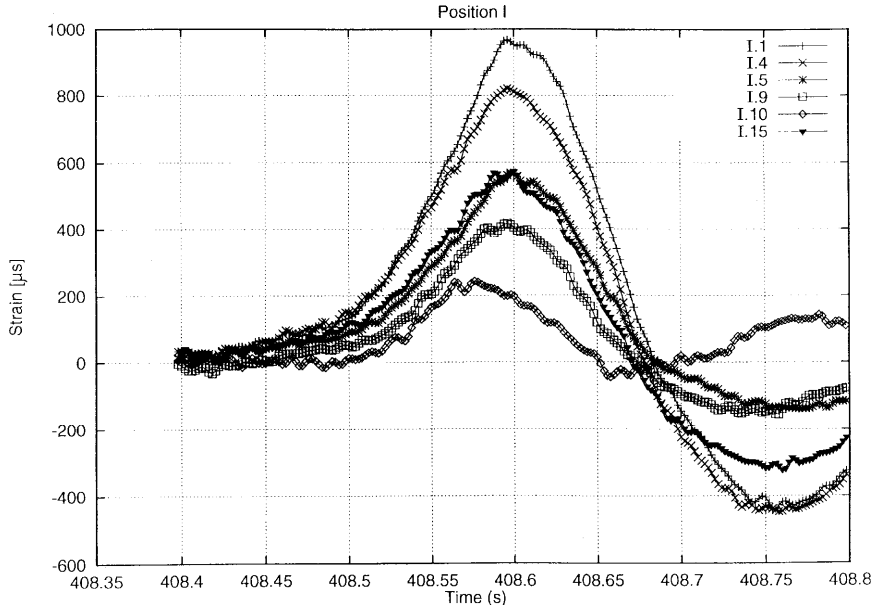


Fig. 17. Measured strains at position I of the Ulstein test vessel. Test No. 204. $H_{1/3} = 1.5$ m; $T_p = 6$ s; head sea; $U = 18$ knots³⁴

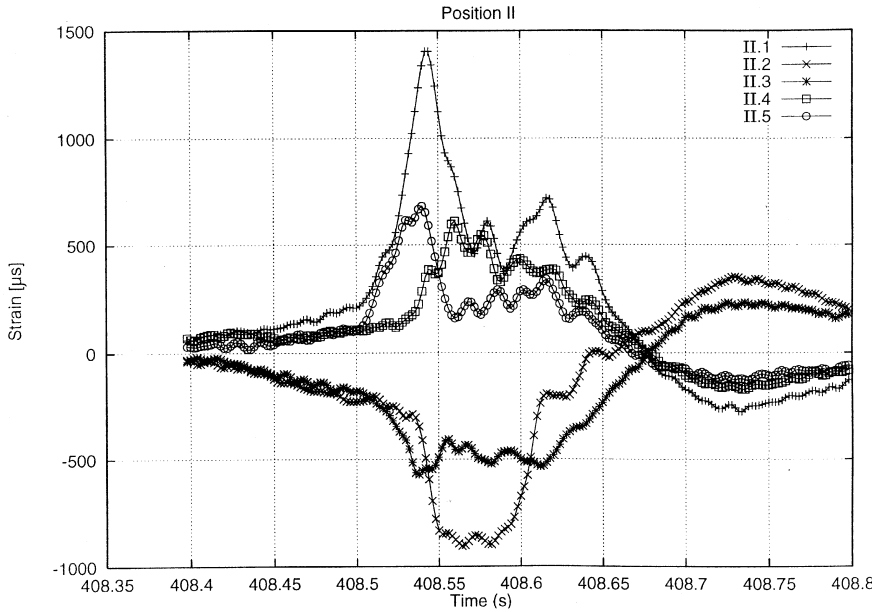


Fig. 18. Measured strains at position II of the Ulstein test vessel. Test No. 204. $H_{1/3} = 1.5$ m; $T_p = 6$ s; head sea; $U = 18$ knots³⁴

tant effect on the maximum strain. This is due to the reduction of the water entry velocity with time.

The estimation of V_0 and V_1 on the impact velocity V (see Eq. 9) is associated with uncertainties. It is therefore appropriate to estimate the errors in maximum strain at position II.1 due to errors in V_0 and V_1 . The hydroelastic theory gave $\partial \epsilon_m / \partial V_0 = 3262 \mu\text{s/ms}^{-1}$ and $\partial \epsilon_m / \partial V_1 = 99 \mu\text{s/ms}^{-2}$. $\partial \epsilon_m / \partial V_0$ was obtained by calculating ϵ_m for $V_0 \pm 0.05$ m/s with V_1 constant. $\partial \epsilon_m / \partial V_1$ was found by calculating ϵ_m for $V_1 \pm 0.5$ m/s² with V_0 constant. The uncertainty analysis shows that the results are particularly sensitive to V_0 , but V_1 plays an important role in

the results since it reduced the water entry velocity when maximum strain occurs.

The hydroelastic orthotropic plate theory is an approximation. The error sources are of both a structural and a hydrodynamic nature. The structural model has been verified by comparing it with finite-element calculations. A quasisteady model was used. When the water has passed the position of the middle longitudinal stiffener, and the strain at position II.1 has reached its maximum value, the agreement between orthotropic plate theory and the finite-element method is good. When the strain at position II.5 has its maximum value,

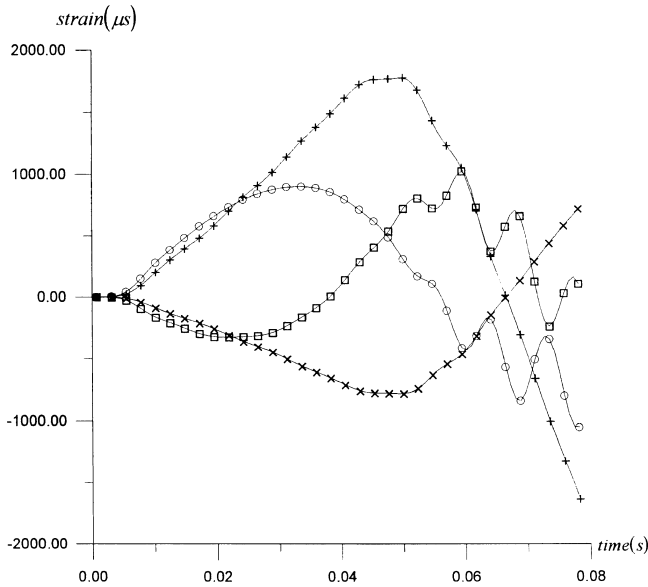


Fig. 19. Theoretical strains based on hydroelastic orthotropic plate theory at position II of the Ulstein test vessel. Test No. 204. $H_{1/3} = 1.5$ m; $T_p = 6$ s; head sea; $U = 18$ knots; $V_0 = 2.6$ m/s; $V_1 = -25.8$ m/s². Upright crosses, II.1; diagonal crosses, II.2 and 3; squares, II.4; circles, II.5

the agreement is less good. This occurs when a small part of the plate is wetted. Results using the orthotropic plate theory are not sensitive to the number of modes if at least two modes are used in the y -direction.

Hydrodynamic errors occur because the body boundary and free surface conditions are approximated. Further, the quadratic velocity term in Bernoulli's equation is neglected in calculating the pressure. These effects can be assessed by comparing them with the results of Zhao and Faltinsen.³⁵ They presented numerical results for water entry by rigid wedges with a constant velocity. Gravity was neglected and incompressible potential flow was assumed. When the dead-rise angle is 15°, Zhao and Faltinsen³⁵ predict a vertical water entry force $F_3 = 85.5\rho V^3 t$. Our theoretical approach gives $108\rho V^3 t$ for rigid-wedge impact. This gives an indication of the errors in strains which arise from an approximation of Bernoulli's equation, body boundary, and free-surface conditions. Since the differences in loading by these methods are clearly largest at the spray roots, the resulting difference in the two forces is not a conservative estimate of error. The error will decrease with decreasing dead-rise angle. The effects of gravity, compressibility, and air cushions are believed to be small (see the discussion in the introduction). Hydrodynamic interaction effects with the side-hulls will be present. These have been discussed earlier. The hydroelastic boundary value problem has been approximated. A mean deflection velocity has been used in formulating the body boundary condition. A more complete formula-

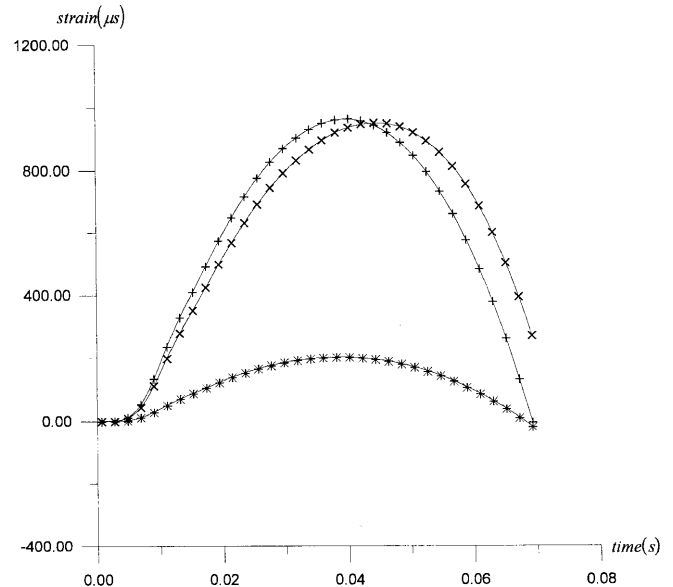


Fig. 20. Theoretical strains based on hydroelastic orthotropic plate theory at position I of the Ulstein test vessel. Test No. 204. $H_{1/3} = 1.5$ m; $T_p = 6$ s; head sea; $U = 18$ knots; $V_0 = 1.75$ m/s; $V_1 = -17.7$ m/s². Upright crosses, I.1; diagonal crosses, I.4; stars, I.5

tion is needed to assess the error, but studies by Kvålsvold and Faltinsen²¹ indicate that this is not a major error source.

Figure 20 shows theoretical predictions at impact position I. These agree reasonably well with experimental results at I.1 and I.4. The agreement is not good for I.5. The same is true for I.15. One error source in the theoretical predictions is the end conditions at the transverse stiffeners. The theory assumes zero deflections. This is not consistent with the measured strains at position I.9 on the transverse stiffener at frame 25 (see Fig. 17). In fact, the measured strains at I.9 are comparable with the measured strains at I.5 and I.15. The measured strains in the transverse stiffeners at position II were much more consistent with the theoretical assumptions. Care was taken that sufficient modes were used in the theoretical calculations. This was done by a quasisteady analysis. This indicates a relative error of around 5%. There are also errors associated with orthotropic plate theory in the same way as was discussed for position II. Since the dead-rise angle at position I is much smaller than at position II, the hydrodynamic errors associated with the approximation of the Bernoulli's equation and the body boundary and free-surface conditions would be less at position I than at position II. The error due to the hydrodynamic interaction with the side-hulls is less at position I than at position II. Figure 17 shows that the maximum strain at I.10 is much lower than that at I.1. One reason is that the distance between frames 25 and 26 is 0.92 m, while it is 1.25 m between frames 24 and 25.

Beam theory shows that the strains are proportional to the square of that distance. Another reason is the 3-D flow effect. The wet-deck has a different longitudinal heel angle in front of frame 26 than aft of frame 26. This means that frame 26 causes a hydrodynamic 3-D end effect that lower the loads (see discussion in Zhao et al.³⁶). Since the distance from the center line to I.10 is almost the same as the distance from I.10 to frame 26, this 3-D effect is important. Both the theory and the experiments show very small strains at I.2 and I.3. Theoretically this is a consequence of the mode shape in the longitudinal direction. The measured strains at I.11 and I.13 also show very low values. The reason must be a shielding effect by the anchor box on the water impact at I.11 and I.13. Hydroelasticity mattered, but did not have a dominant role in the comparative study with full-scale measurements. Its importance will increase with decreasing dead-rise angle and increasing impact velocity. Significant local hydroelastic effects were only recorded once during the full-scale measurements. These results are shown in Fig. 21, and are strains measured at

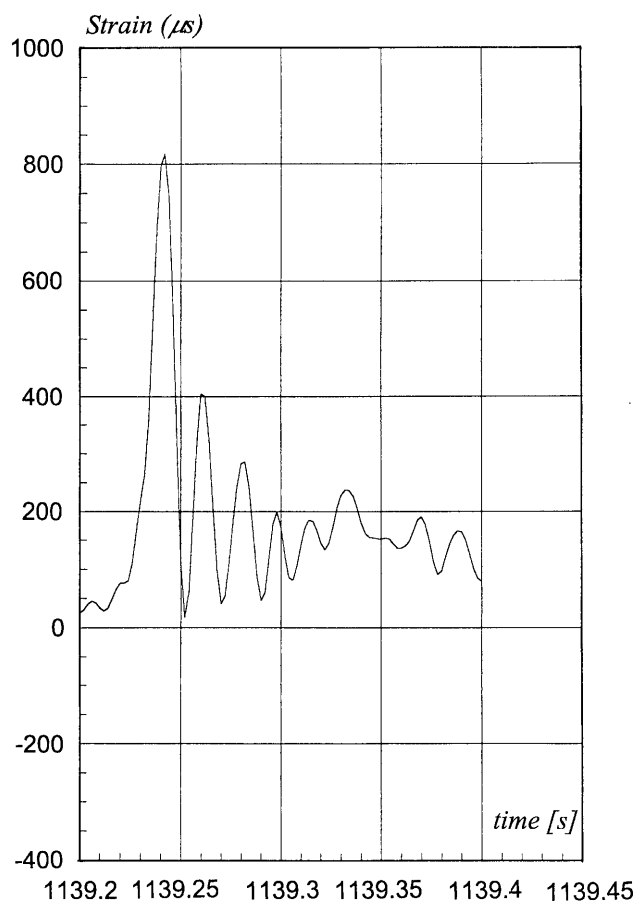


Fig. 21. Measured hydroelastic behavior of the strain response (in μs) at position II.6 of the Ulstein test vessel. Test No. 207. $H_{1/3} = 1.8$ m; $T_p = 5.4$ s; head sea; $U = 6$ knots³⁴

position II.6 during test no. 207. This means that $H_{1/3} = 1.8$ m and $T_p = 5.4$ s in head-sea conditions. The vessel speed was 6 knots. No other strain gauges showed significant local hydroelastic effects. This means that the flow was asymmetric about the centerplane. This may be due to rolling and/or interacting water motion between the two hulls. The latter is possible since the forward speed is low. It will not be dominant at position I due to three-dimensional hydrodynamic end effects and the nonzero forward speed. It is more likely to occur at position II. Even if the main wave direction was a head sea, relatively large roll amplitudes were recorded. The reported roll standard deviation and maximum amplitude during 220 roll oscillations were 1.44° and 5.7° . The first half of the oscillation period is about 0.025 s in Fig. 21. This agrees with the theoretical estimate. The subsequent three oscillations have a period less than 0.02 s but higher than 0.01 s, which is the measured dry natural period. A possible explanation is presence of air bubbles after the first half of the oscillation period. The reason why the amplitude drops after the first half of the oscillation period is the decrease in the natural period. The dynamic behavior has similarities with the dynamic behavior during the drop tests described earlier for the one-beam model (see Fig. 7). However, the physical reason for the change in period was then cavitation and ventilation. Since we have no means of estimating the impact velocity V and the relative angle β between the water and the body at impact, we cannot make a comparison with the theory.

3 Conclusions

An overview of the many impact problems occurring in ship and ocean engineering is given. Many physical effects occur during slamming, but it is emphasized that the slamming problem must be hydrodynamically analyzed from a structural point of view. This implies, for instance, that compressibility and air cushions may have small effects.

Nonlinearities in the waves and interaction with the structure can be important in describing the impacting water surface. Some examples are wet-deck slamming, green water on deck, and sloshing in tanks.

When the angle between the impacting free surface and the body surface is small, hydroelasticity has to be considered. The importance of hydroelasticity will also depend on the impact velocity V and the highest natural period (T_{n1}) of the local structure. The higher the values of V and T_{n1} , the more important is hydroelasticity. When hydroelasticity is important, the introduction of a design pressure has no physical meaning. One should directly study maximum strains as a consequence of the impacting free surface. This is illustrated by the results

Table 4. Results from the parameter studies

Parameter	Important	Not important
Initial impact position		X
Structural mass, M_B		X
Bending stiffness, EI	X	
Beam length L_B	X	
Radius of curvature R of impacting free surface		X
Relative angle α_{rel} (see Eq. 2)	X	
Relative impact velocity V_T (see Eq. 1)	X	
Time history of velocity $V_T(t)$	X	
Side hulls	X	

of drop tests of horizontal or nearly horizontal elastic plates. Theoretical studies are also reviewed. It is shown that the forward speed can be handled as an angle-of-attack effect.

Comparative studies between theoretical predictions and full-scale experiments of local slamming-induced strains in the wet-deck of a 30-m-long catamaran, reported by Faltinsen,³² are presented. Fair agreement between theory and experiment was recorded, but the predicted strains were very sensitive to the time-varying impact velocity. The impact velocity was not measured, and is strongly dependent on the nonlinear hull and free-surface effect. Current state-of-the-art computer programs cannot predict the impact velocity sufficiently accurately. The most important error source in the theoretical analysis is the impact velocity. The wet-deck of the 30-m catamaran being considered had a wedge-shaped cross-sectional form with a dead-rise angle of 14° in the initial impact area. Hydroelasticity is then insignificant. The measured maximum strains correspond to about half the yield stress. This occurred in a head sea with a significant wave height, $H_{1/3} = 1.5$ m, and a ship speed of 18 knots. The ship was allowed to operate up to $H_{1/3} = 3.5$ m.

A summary of the important parameters for hydroelastic impact on steel and aluminum structures is given in Table 4. The comment on side-hull effects refers to wet-deck slamming on a catamaran.

References

- Korobkin A (1996) Water impact problems in ship hydrodynamics. In: Ohkusu M (ed) *Advances in marine hydrodynamics*, chap 7. Computational Mechanics Publications, Southampton, Boston
- Faltinsen OM (2000) Water impact in ship and ocean engineering. Keynote Lecture. In: Goda Y, Ikehata M, Suzuki K (eds) *Proceedings of the 4th International Conference on Hydrodynamics (ICHHD 2000)*. ICHD 2000 Local Organizing Committee, Yokohama, pp 17–36
- Buchner B (1995) On the impact of green water loading on ship and offshore unit design. In: Kim H, Lee JW (eds) *Proceedings PRADS'95*, Seoul, Korea. Society of Naval Architects of Korea, pp 1430–1443
- Greco M, Faltinsen OM, Landrini M (2000) Basic studies of water on deck. In: Miloh, T, Zilman G (eds) *Proceedings of the 23th Symposium on Naval Hydrodynamics*, Val de Reuil, France. National Academy Press, Washington D.C.
- Faltinsen OM (2000) Slamming on ships. Keynote lecture, IMAM 2000, Naples. Navale Università di Napoli
- Faltinsen OM, Rognebakke OF (1999) Sloshing and slamming in tanks. *Hydronav'99–Manoeuvring'99*, Gdansk–Ostrada, Poland.
- Faltinsen OM, Rognebakke OF (2000) Sloshing. Keynote Lecture, NAV2000, Venice. Cetena, Genova
- Rognebakke OF, Faltinsen OM (2000) Damping of sloshing due to tank roof impact. 15th International Workshop on Water Waves and Floating Bodies, Caesarea, Israel. Tel Owin University, Israel
- Faltinsen OM, Rognebakke OF, Lukovsky JA, Timokha AM (2000) Multidimensional modal analysis of non-linear sloshing in a rectangular tank with finite water depth. *J Fluid Mech* 407:201–234
- Faltinsen OM, Timokha AM (2000) Adaptive multimodal approach to nonlinear sloshing in a rectangular tank. *J Fluid Mech*, in press
- Ulstein T, Faltinsen OM (1994) Hydroelasticity analysis of a flexible bag structure. *Proceedings of the 20th Symposium on Naval Hydrodynamics*, University of California, Santa Barbara. National Academy Press, Washington D.C.
- Ulstein T, Faltinsen OM (1998) Cobblestone effect on SES. Applied Vehicle Technology Panel. Symposium on Fluid Dynamics Problems of Vehicles Operating near or in the Air–Sea Interface, Amsterdam. (Research and Technology Agency, NATO) Neuilly-Sur-Seine Cedex, France
- Ulstein T (1995) Nonlinear effects of a flexible stern seal bag on cobblestone oscillations of a SES. Thesis, Department of Marine Hydrodynamics, Norwegian University of Science and Technology, MTA Report 106
- Faltinsen OM (1998) Hydroelasticity of high-speed vessels. Keynote Lecture. In: Kashiwagi M, Koterayama W, Ohkusu M (eds) *Proceedings of the 2nd International Conference on Hydroelasticity in Marine Technology*. RIAM, Kyusku University, Fukuoka, pp 1–13
- Arai M, Miyanchi T (1998) Numerical study of the impact of water on cylindrical shells, considering fluid–structure interactions. PRADS. In: Oosterweld MCW, Tan SE (eds) *Practical design of ships and mobile units*. Elsevier, Amsterdam, pp 59–68 (Developments in Marine Technology, vol 11)
- Kvålsvold J (1994) Hydroelastic modelling of wetdeck slamming on multihull vessels. Thesis, Department of Marine Hydrodynamics, Norwegian Institute of Technology, MTA Report 100
- Baarholm R, Faltinsen OM (2000) Experimental and theoretical studies of wave impact on an idealized platform deck. 4th International Conference on Hydrodynamics (ICHHD 2000). ICHD 2000 Local Organizing Committee, Yokohama, pp 181–186
- Faltinsen OM, Zhao R (1998) Water entry of a wedge into a channel. In: Hermaus AJ (ed) *Proceedings of the 13th International Workshop on Water Waves and Floating Bodies*, Alphen aan den Rijn, The Netherlands. Delft University of Technology
- Faltinsen OM (1997) The effect of hydroelasticity on slamming. *Phil Trans R Soc London, Ser. A* 355:575–591
- Faltinsen OM, Kvålsvold J, Aarsnes JV (1997) Wave impact on a horizontal elastic plate. *J Mar Sci Technol* 2(2):87–100
- Kvålsvold J, Faltinsen OM (1995) Hydroelastic modelling of wetdeck slamming on multihull vessels. *J Ship Res* 39:225–239
- Kvålsvold J, Faltinsen OM, Aarsnes JV (1995) Effect of structural elasticity on slamming against wetdecks of multihull vessels. *Proceedings of PRADS'95*, Korea. Society of Naval Architects of Korea, pp 1684–1699

23. Aarsnes JV (1994) An experimental investigation of the effect of structural elasticity on slamming loads and structural response. Technology Report MT60 A95-0053,602119.00.01, Norwegian Maritime Technology Research Institute A/S, MARINTEK
24. Korobkin AA, Khabakhpasheva TI (1999) Periodic water impact onto an elastic plate. In: Piquel J (ed) Proceedings of the 7th International Conference on Numerical Ship Hydrodynamics, Nantes. Ecole Centrale de Nantes
25. Miyamoto T, Tanizawa K (1985) A study of the impact on ship bow (2nd report). J Soc Nav Archit Jpn 158:270–279
26. Wagner H (1932) Über Stoss- und Gleitvorgänge an der Oberfläche von Flüssigkeiten. Z Angew Math Mech 12:193–235
27. Faltinsen OM (1990) Sea loads on ships and offshore structures. Cambridge University Press, Cambridge
28. Haugen EM (1999) Hydroelastic analysis of slamming on stiffened plates with application to catamaran wetdeck. Thesis, Department of Marine Hydrodynamics, Norwegian University of Science and Technology
29. Haugen EM, Faltinsen OM (1999) Theoretical studies of wetdeck slamming and comparisons with fullscale measurements. Proceedings of the 5th International Conference on Fast Sea Transportation (FAST'99), Seattle. SNAME, Jersey City, New Jersey
30. Haugen EM, Faltinsen OM, Aarsnes JV (1997) Application of theoretical and experimental studies of wave impact to wetdeck slamming. 4th International Conference on Fast Sea Transportation (FAST'97), Sydney. Publisher: Baird Publications, South Yarra Victoria, pp 423–430
31. Aarsnes JV (1996) Drop test with elastic 3-beam model. Effect of forward speed. Technical Report MT60 F96-0291, 603029, Norwegian Maritime Technology Research Institute A/S, MARINTEK
32. Faltinsen OM (1999) Water entry of a wedge by hydroelastic orthotropic plate theory. J Ship Res 43:180–193
33. Szilard R (1974) Theory and analysis of plates. Classical and numerical methods. Prentice-Hall, Englewood Cliffs
34. Aarsnes JV, Hoff RJ (1998) Full scale with Ulstein test vessel. Technocal Report MT60 F98-205, 603852.00.01, Norwegian Maritime Technology Research Institute A/S, MARINTEK
35. Zhao R, Faltinsen OM (1993) Water entry of two-dimensional bodies. J Fluid Mech 246:593–612
36. Zhao R, Faltinsen OM, Aarsnes JV (1996) Water entry of arbitrary two-dimensional sections with and without flow separation. Proceedings of the 21st Symposium on Naval Hydrodynamics, Trondheim. National Academy Press, Washington DC, pp 408–423